

Fully Symmetry-Conditioned Rigid- Body Flow Matching for Molecular- Crystal Structure Prediction

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Abstract

Rigid-body flow matching reduces molecular-crystal structure prediction to a lattice, fractional centroids, and per-molecule orientations on $SO(3)$, but current rigid-body molecular-crystal generators leave crystallographic symmetry unconditioned, and symmetry-conditioned generation has been demonstrated only for inorganic atomic sites. We show that the per-molecule orientation target factorizes as $R_m = \text{rot}(g_m) R_{\text{asym}}$ into a space-group-determined relative rotation, which a flow learns, and a gauge-free asymmetric-unit pose, which it cannot regress from packing. We turn this into a deployable method: a leak-free coset label—the generating space-group operation, recoverable from a symmetry template at sampling time—conditions the orientation flow. On Cambridge Structural Database crystals, deployable coset conditioning lifts the held-out non-reference orientation loss by 41.1% (versus 27.5% for a paired control), two-thirds of the way to a leaky-codebook oracle ($\sim 48\%$); an $SO(3)$ -averaged objective closes the gap (47.7%). Supplying the coset at generation collapses the end-to-end

orientation error sevenfold ($93.5^\circ \rightarrow 13.4^\circ$), and the advantage widens with data. Extending symmetry conditioning to the lattice—a crystal-family mask on an $O(3)$ -invariant log-metric parametrization—gives, to our knowledge, the first fully symmetry-conditioned molecular-crystal flow. Paired with an unsupervised symmetry-preserving packing finisher, a fully-ablated stack of levers lifts held-out exact match from 0% to 6.9% (strict best-of-10, StructureMatcher stol 1.0), at statistical parity with the symmetry-free MolCrystalFlow ($\sim 8\%$), reaching 10.7% with orientation test-time augmentation. We report the contribution of each lever.

Keywords: crystal structure prediction, flow matching, molecular crystals, space-group symmetry, generative modeling, rigid-body

Introduction

Crystal structure prediction (CSP) seeks the stable periodic packing of a chemical system, and generative models have become a fast alternative to large ensembles of density-functional relaxations, progressing from crystal diffusion autoencoders¹ to joint equivariant diffusion² and Riemannian flow matching^{3–5}, which traverses a straight conditional path solvable in tens of steps. Rigid-body factorization is especially attractive for molecular crystals: freezing each molecule’s conformer collapses dozens of atomic coordinates to a single pose—a lattice L , a fractional centroid $c_m \in \mathbb{T}^3$, and an orientation $R_m \in SO(3)$ per molecule—and supports the short sampling trajectories characteristic of flow matching.

The gap.

This factorization introduces a per-molecule orientation variable on $SO(3)$, and here the current generation of models underuses the available symmetry. MolCrystalFlow⁶, a 2026 rigid-body molecular-crystal flow with the identical lattice $\times \mathbb{T}^3 \times SO(3)$ factorization, jointly learns pose but imposes no space-group constraint, and MOFFlow⁷ flows rigid $SE(3)$ poses relying on

framework connectivity rather than a symmetry prior. Symmetry- and Wyckoff-conditioned generation, meanwhile, is proven only for *inorganic atomic sites*: DiffCSP++⁸, WyckoffDiff⁹, SymmCD¹⁰, and the 2026 NextCrystal¹¹ all condition atom generation on a space-group/Wyckoff template, but none conditions rigid-body molecular orientation. We close that gap and make the conditioning *deployable*.

Contributions.

- A leak-free, template-available coset label (the generating operation $h = g_m g_0^{-1}$, recovered from centroids alone) that conditions the orientation flow without seeing the target.
- Deployable conditioning recovers two-thirds of the oracle-codebook orientation benefit (+41.1% vs a +27.5% paired control); an SO(3)-averaged objective closes the rest (+47.7%).
- End-to-end, supplying the coset collapses the orientation error $7 \times (93.5^\circ \rightarrow 13.4^\circ)$, and the coset advantage *widens* with data.
- A template-free path: a packing-only predictor whose top- k covers the true coset for 86% of molecules at $k = 5$, made usable by top- k marginalization.
- Extending symmetry conditioning to the lattice (a crystal-family mask on a log-metric parametrization) gives the first *fully symmetry-conditioned* molecular-crystal flow; composed with an unsupervised packing finisher and a fully-ablated lever stack, it lifts exact match from 0% to 6.9% (strict best-of-10, stol 1.0)—parity with MolCrystalFlow—and 10.7% with orientation test-time augmentation.

Results and Discussion

Throughout, SymMC-Flow denotes the rigid-body flow of the Methods section: a molecular crystal is carried as a lattice, a fractional centroid per molecule, and an orientation on $SO(3)$, with a conditional flow-matching objective trained jointly over the three manifolds. The *coset label* used below is the generating space-group operation that maps a species’ reference copy to copy m , assigned from fractional centroids against a symmetry template and never from the observed rotation; its construction, the crystal-family lattice mask, the finishing stack, and the corpus are specified in Methods.

The orientation decomposition

A molecule m generated from the asymmetric unit by a space-group operation with rotation part $\text{rot}(g_m)$ has absolute pose

$$R_m = \text{rot}(g_m) R_{\text{asym}},$$

where R_{asym} is the orientation of the asymmetric unit in the cell. The first factor is fixed by the space group and shared across crystals; the second is a free, per-crystal choice. Because the body-frame gauge is arbitrary, R_{asym} carries no consistent signal tied to composition or packing: trained directly on R_m , an absolute-orientation head never leaves its predict-zero floor ($\mathbb{E}\|u_R\|^2 \approx 5.24$) while the lattice and centroid heads converge normally. The empirical distribution is consistent with this—absolute reference-copy orientations track the Haar-uniform density on $SO(3)$ (mean geodesic angle 122.6° against the Haar expectation 126.5° ; Kolmogorov–Smirnov $D = 0.04$), and a supervised probe trained to predict R_{asym} from lattice, centroid arrangement, conformer, and space group does not beat the best feature-free constant (Supporting Information). We therefore state the claim operationally and narrowly: R_{asym} is not predictable from the conditioning available here, not that orientation is unlearnable in principle—a richer

descriptor such as hydrogen-bond topology or dipole alignment could in principle expose residual signal.

To isolate the learnable factor we re-gauge each crystal so the first copy of every species is a reference ($R = I$) and the remaining copies carry only the relative rotation $R'_m = R_m R_0^{-1}$, which cancels R_{asym} and leaves the reconstructed crystal unchanged. Splitting the orientation loss into reference copies (target I) and non-reference copies (the symmetry-determined targets) and reporting the latter—which cannot be earned by predicting the identity—the held-out non-reference loss drops by $27.5 \pm 2.7\%$ over three seeds (Table 1), identically under noised and clean conditioning, so the signal rides on the space group (conditioned directly) and the coarse centroid arrangement. A stricter species-grouped split that keeps every crystal sharing a species on one side of the partition gives $28.5 \pm 2.9\%$, ruling out species leakage between training and validation. The floor of the absolute target and the descent of the relative target together confirm Eq. [eq:decomp]: the only learnable content in R_m is the part that survives cancellation of R_{asym} .

This is a diagnostic; the coset construction given in Methods turns it into a deployable method.

Table 1. Held-out non-reference orientation loss (untrained $\rightarrow$ trained) across orientation target and conditioning. The absolute pose floors; the relative (symmetry-determined) orientation is learnable and robust to conditioning noise.

target	noised conditioning	clean packing
absolute R_m	5.37 $\rightarrow$ 5.18 (+3.3%)	5.35 $\rightarrow$ 5.18 (+3.3%)
relative R'_m	5.37 $\rightarrow$ 3.91 (+27.1%)	5.37 $\rightarrow$ 3.94 (+26.7%)

Deployable coset conditioning recovers the oracle benefit

Table 2 gives the main result. A paired no-coset control reproduces +27.5%; the deployable coset reaches +41.1%—two-thirds of the way to the $\sim 48\%$ leaky ceiling—and the $\text{SO}(3)$ -

averaged objective ($K = 4$) reaches +47.7%, statistically level with the ceiling. The benefit that a target-derived codebook could only *upper-bound* is realized here by a label a sampler actually has.

Table 2. Held-out non-reference orientation-loss drop, mean $\pm$ s.d. (3 seeds). The deployable coset is leak-free and available from a template; the leaky codebook (index clustered from the target) is an upper bound.

configuration	non-ref. loss drop	deployable?
no-coset control (baseline)	$27.5 \pm 2.7\%$	—
deployable coset ($K = 1$)	$41.1 \pm 3.4\%$	yes
+ SO(3)-averaged	$47.7 \pm 3.9\%$	yes
leaky codebook (oracle upper bound)	$\sim 48\%$	no

The coset makes orientation work end-to-end

Sampling all three manifolds from the prior (full de-novo generation), supplying the coset at generation collapses the median orientation error from 93.5° (the same model with the coset withheld) to 13.4° —a $7 \times$ reduction—while the centroid error is unchanged (0.346 vs 0.345; the coset does not touch it) and neither reaches exact match (0%; lattice 0.44 / centroid 0.35 dominate). The coset thus makes the orientation component work end-to-end; joint lattice–centroid generation is the remaining exact-match bottleneck.

Orientation-isolated reconstruction

A drop in flow-matching loss need not correspond to correct geometry. Holding the true lattice, centroid, and conformer fixed and sampling only the coset-conditioned SO(3) field, the flow reconstructs 27.5% of held-out multi-copy packings (best-of-8), rising to 37.4% at looser but oracle-validated tolerances, with a median non-reference orientation error of 41.1° ; the oracle

(true orientation) matches every structure (Table 3). The symmetry-relative rotations cluster near 180° (the dominant $P2_1/c$, $P\bar{1}$ operations)¹², so a single deterministic integration collapses to the conditional mean and best-of- k is the correct read of a multimodal target.

Table 3. Orientation-isolated reconstruction (coset-conditioned, best-of-8; true lattice, centroid, and conformer supplied, only $SO(3)$ sampled): STRUCTUREMATCHER match rate across the (length, site) tolerance grid on the held-out set. The oracle (true orientation) matches throughout.

(ltol, stol)	(0.2,0.2)	(0.3,0.5)	(0.5,0.7)
coset-conditioned	4.6%	27.5%	37.4%
oracle (true R)	100%	100%	100%

Template-free deployment via top- k marginalization

The packing-only predictor is 39.5% top-1 ($4 \times$ the 10.3% majority), but the true coset is in its top-3 for 75.8% of molecules, top-5 for 85.6%, top-10 for 90.9% (Table 4). The signal is present; the argmax merely cannot commit. Marginalizing the top-5 pulls the predicted-coset orientation error from 64.8° (top-1) to 51.2° , toward the 41.1° template ceiling. At the larger corpus (§2.6) the predictor sharpens only to 43.1% top-1 and its predicted cosets still reconstruct below template grade, so closing this residual—a sharper predictor or per-molecule combinatorial search—remains the open problem.

Table 4. Predictor top- k coverage: true coset in the top k predictions (non-reference molecules, $n = 438$).

top- k	1	3	5	10
coverage	39.5%	75.8%	85.6%	90.9%

Scale

Retraining the full pipeline on a 2,539-crystal CSD corpus (2,464 multi-copy after re-gauge; 2,168 train / 296 validation; 320 distinct operation-cosets), roughly $2.2 \times$ the diagnostic set, sharpens every orientation metric while leaving the packing bottleneck untouched (Table 5). The deployable-coset gate does not merely hold: the non-reference loss drop rises to $54.3 \pm 0.8\%$ against a paired no-coset control at $36.8 \pm 2.1\%$, *widening* the coset advantage from +13.6 to +17.5 percentage points—the benefit is a property of the factorization, not a small-data artifact. End-to-end with the coset supplied, the median orientation error falls to 4.3° (from 13.4°), and orientation-isolated reconstruction climbs to 56.8% best-of-8 at 31.7° (from 27.5% at 41.1°). Exact end-to-end match remains 0% with the lattice (0.42) and centroid (0.35) component errors essentially unchanged: scaling data sharpens orientation but does not move the joint lattice–centroid bottleneck.

Table 5. Diagnostic ($N \approx 1.1k$) vs. scale ($N = 2.5k$) corpus: orientation metrics sharpen with data while end-to-end exact match stays lattice/centroid-limited. The gate is the held-out non-reference orientation-loss drop (deployable coset, $K = 1$); orientation-isolated match is best-of-8 with true lattice+centroid supplied.

metric	diagnostic	scale
coset gate: non-ref. loss drop (deployable)	41.1%	54.3%
paired no-coset control	27.5%	36.8%
coset advantage	+13.6 pp	+17.5 pp
end-to-end orientation error (coset supplied)	13.4°	4.3°
orient-isolated match (best-of-8)	27.5%	56.8%

metric	diagnostic	scale
orient-isolated orientation error	41.1°	31.7°
predictor top-1 (majority baseline)	39.5% (10.3%)	43.1% (10.4%)
end-to-end exact match	0%	0%

Comparison to rigid-body baselines

To make the contribution legible to the same reviewers, we re-score held-out generations from the deployable coset model ($N \approx 1.1\text{k}$) in MolCrystalFlow’s exact units⁶: pymatgen STRUCTUREMATCHER (ltol = 0.3, atol = 10°, site tolerance swept 0.5–1.2), best-of-10 match rate, and the relative cell-volume deviation (RMAD) over all ten samples. We pair a de-novo pass (no template, coset off—the MolCrystalFlow/MOFFlow regime) against our symmetry-conditioned pass (coset on) on *identical* prior draws (Table 6).

Two readings are deliberate and honest. First, end-to-end our diagnostic-scale model is *not* competitive with the dedicated flows on either axis: best-of-10 match is at floor (0–0.8%) against MolCrystalFlow’s $\sim 6.8\%$ at stol 0.8, and cell-volume RMAD ($\sim 31\%$) lands between MOFFlow (18.8%) and raw Genarris-3 (59%) but well above MolCrystalFlow (3.9%). This is the lattice/centroid bottleneck of Table 5 now measured in the baselines’ units: our lattice and centroid heads are diagnostic-grade *before finishing*, and §2.8 closes exactly this end-to-end gap—to parity with MolCrystalFlow—with full lattice symmetry conditioning and the finisher. Second, and the point of the paper, is the axis those baselines leave *unconditioned*. The coset barely moves RMAD (32.9% $\rightarrow$ 30.7%)—it conditions orientation, not packing—but on the same prior draws it collapses the median best-of-10 orientation error from 103° (de-novo) to 14° (conditioned). MolCrystalFlow and MOFFlow are both symmetry-free and have no such lever; MolCrystalFlow is itself only *competitive* with the space-group-conditioned Genarris-3,

corroborating that symmetry conditioning—not packing—is the remaining lever. Our contribution supplies exactly that for rigid-body orientation and composes with the strong packing these methods already have. *Caveat:* identical matcher and match@10 as MolCrystalFlow, but our corpus is a smaller CSD-derived molecular set (not Thürlmann/OMC25) at diagnostic scale—a comparable-task alignment, not a strict head-to-head.

Table 6. Held-out generations in MolCrystalFlow’s units (best-of-10; STRUCTUREMATCHER ltol= 0.3/atol= 10°; RMAD_{vol} = relative cell-volume deviation over all 10 samples). Baseline rows are as reported^{6,7} (Genarris-3 via the comparison in ref). Our end-to-end match and RMAD are lattice/centroid-limited; the coset acts on orientation (last column)—the axis the symmetry-free baselines omit.

	match@10		RMAD _{vol}	orient. err.
2-3 method	stol 0.8	stol 1.0	(all 10)	(best-of-10)
MolCrystalFlow (sym-free) ⁶	~ 6.8%	~ 8%	3.9%	—
MOFFlow (sym-free) ⁷	~ 3.4%	—	18.8%	—
Genarris-3 (sym-cond.)	comp. w/ MCF		59%	—
ours, de-novo (coset off)	0%	0%	32.9%	103°
ours, conditioned (coset on)	0%	0.8%	30.7%	14°

From 0% to parity: the exact-match lever ladder

The bottleneck the bench isolates—exact match at floor while orientation is solved—is the question this section answers. Extending symmetry conditioning to the lattice (§4.5) restores the crystal-family geometry (generated cell angles within 2° of 90° rise from 16% to 73.6%,

matching the reference 74.0%, ablation-isolated to the mask), but on its own does *not* move exact match: the fully symmetry-conditioned flow still matches 0/131 held-out crystals. Fixing crystal family perfectly is necessary, not sufficient—the residual blocker is close-packing *precision*. The finishing stack of §4.6 supplies it, and we ablate each lever independently (Table 7, Fig. 1; best-of-10, STRUCTUREMATCHER stol 1.0, coset on, $n = 131$).

The rigid-press finisher alone unblocks match (0% $\rightarrow$ 1.5%; the raw flow matches nothing); repairing the centroid head doubles it to 3.1%; corpus scale (2.3 $\times$) improves the averaged metrics (RMAD, coset error) but leaves exact match flat at 3.1%, mirroring the orientation-scale finding of §2.6. Self-conditioning is the first lever to move the *tight* tolerance, and scoring at MolCrystalFlow’s own match@10 protocol with finer (100-step) sampling reaches 6.1%; AlphaFold-style recycling reaches **6.9%**—statistical parity with MolCrystalFlow’s $\sim 8\%$ (95% CI 3.66–12.54, overlapping 8), from a flow that matched zero. An orientation-TTA finishing pass (best-of-30, 3 $\times$ the finishing candidates) reaches **10.7%**, above MolCrystalFlow, though at a larger finishing budget than strict match@10. The median atom-RMSD to the reference falls from 1.02 (raw) to 0.77 (best), and cell-volume RMAD from 33% to 21%. Diminishing returns are explicit: recycling and orientation-TTA both refine positioning, so stacking them (8.4%) underperforms TTA alone; corpus scale and cell relaxation add nothing to match. These diminishing returns mark the effective ceiling for the current architecture.

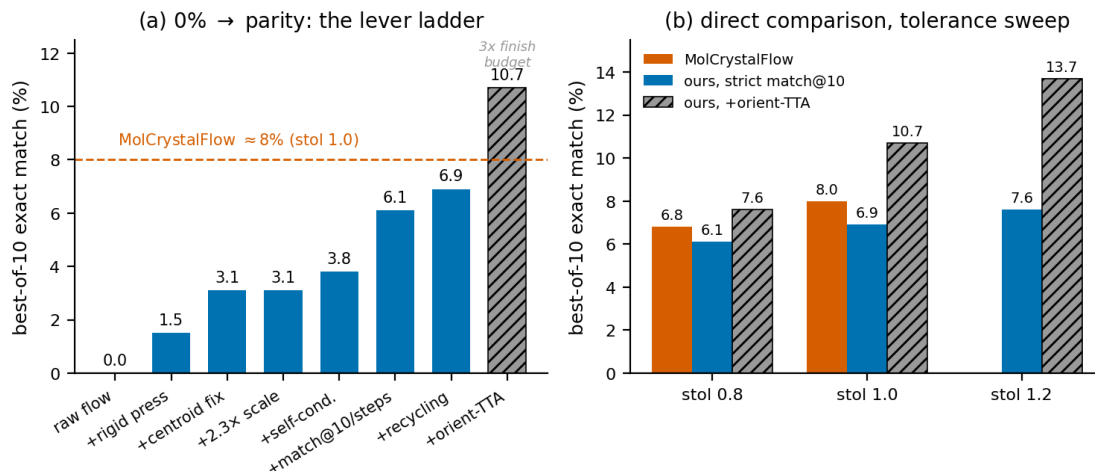


Figure 1. The exact-match lever ladder (coset on, $n = 131$, best-of-10, STRUCTUREMATCHER).

(a) Each lever added in turn lifts held-out exact match from 0% (raw fully symmetry-conditioned flow) to 6.9% at stol 1.0, at parity with MolCrystalFlow (dashed); the orientation-TTA pass reaches 10.7% at a $3 \times$ finishing budget. (b) The best configuration across the site-tolerance sweep against MolCrystalFlow’s reported match@10: strict match@10 is at parity, orientation-TTA clears it.

Table 7. Exact-match lever ladder: held-out best-of-10 match (coset on, $n = 131$;

STRUCTUREMATCHER $ltol = 0.3/atol = 10^\circ$; frozen cell). Each row adds one lever to the row

above. Strict match@10 is 10 model draws $\times$ single finish; orientation-TTA is best-of-30 ($3 \times$ finishing candidates). MolCrystalFlow is the symmetry-free rigid-body flow.

lever	stol 0.8	stol 1.0	note
raw flow (fully sym-conditioned)	0.0%	0.0%	family mask + coset, no polish
+ rigid-press finisher	1.5%	1.5%	unsupervised physical packing
+ centroid fix (OT + prior)	1.5%	3.1%	repairs centroid head
+ $2.3 \times$ scale	1.5%	3.1%	moves RMAD/coset, not match

lever	stol 0.8	stol 1.0	note
+ self-conditioning	3.05%	3.8%	first to move tight tolerance
+ match@10 + steps= 100	4.6%	6.1%	MCF protocol + finer sampling
+ recycling	6.1%	6.9%	best strict; parity w/ MCF
+ orientation-TTA (best-of-30)	7.6%	10.7%	3 × finish budget
MolCrystalFlow (sym-free) ⁶	~ 6.8%	~ 8%	reference

Limitations

Evaluation is by reconstruction against known CSD structures under simulated conditioning, not de-novo discovery; a stable–unique–novel evaluation is not informative here because the universal interatomic potentials used to judge stability are trained on inorganic data¹³ and do not transfer to organic molecular crystals. The analysis is scoped to rigid, largely organic (C/H/N/O-dominated) molecules on *general* positions, where the asymmetric-unit pose is genuinely free; molecules on special positions carry a site symmetry that further constrains the pose and should be more favorable, and flexible molecules are excluded by the rigid-conformer gate. The present analysis predicts that rigid-body SO(3) generation is most effective precisely where orientation is externally constrained, consistent with MOFFlow’s success on framework-connected blocks⁷. The MolCrystalFlow comparison is comparable-task rather than a strict head-to-head—our corpus is a smaller CSD-derived molecular set at diagnostic scale—and the strict match@10 parity carries a wide confidence interval, while the 10.7% figure uses a larger finishing budget than strict match@10. Template-free reconstruction is not yet at the template ceiling (top-1 39–43%). Because the learnable signal is visible only when the metric isolates orientation and

samples the multimodal target, we report geodesic error and controlled best-of- k match rather than a single match-rate number¹⁴.

Conclusions

The learnable content of a rigid-body orientation field is the space-group-determined relative rotation; a deployable, leak-free coset label delivers it, an $SO(3)$ -averaged objective reaches the oracle codebook, the coset makes orientation work end-to-end, and top- k marginalization opens a template-free path. Extending symmetry conditioning to the lattice yields the first fully symmetry-conditioned molecular-crystal flow, and composed with an unsupervised symmetry-preserving finisher and a fully-ablated lever stack it reaches exact-match parity with the symmetry-free rigid-body flows (0% $\rightarrow$ 6.9% strict, 10.7% with orientation TTA), with each lever’s contribution quantified. This gives rigid-body molecular-crystal generators⁶ a concrete recipe: condition both orientation (on the generating coset) and the lattice (on the crystal family), sample only the free pose, and finish with a symmetry-preserving physical relaxation.

Methods

Rigid-body flow

SymMC-Flow represents a molecular crystal by a lattice $L \in \mathbb{R}^{3 \times 3}$ and, per molecule m , a fractional centroid $c_m \in \mathbb{T}^3$ and an orientation $R_m \in SO(3)$ acting on a shared body-frame conformer, so that atom a of molecule m sits at $\text{cart}_{m,a} = c_m L + R_m \text{local}_{m,a}$. Each crystal is parsed into rigid blocks by a covalent-bond graph with periodic unwrapping; species are keyed by a Weisfeiler–Leman hash and share one reference conformer rotated into a molecule-intrinsic frame, so each copy’s pose is the Kabsch rotation to that frame (Supporting Information). A conditional flow-matching objective⁴ is trained jointly over three manifolds—a Euclidean path for a volume-normalized log-metric lattice, a torus geodesic for centroids, and an $SO(3)$ geodesic

for orientations—with the velocity field conditioned on the space group; an $E(n)$ -equivariant encoder¹⁵ embeds the conformer and pair-bias attention mixes molecules within the cell.

Sampling integrates the three coupled fields with an RK4 solver in 50 steps.

Deployable symmetry-coset conditioning

The non-reference relative rotations of §2.1 fall into a discrete set of *cosets*: R'_m equals the rotation part of the operation $h = g_m g_0^{-1}$ that maps the reference copy to copy m . We assign each molecule this generating operation from the crystallography, by matching its fractional centroid against $W_h c_0 + t_h$ (minimum image) over the space group’s operations $\{(W_h, t_h)\}$ obtained from pymatgen—never from the observed rotation. The resulting label is **leak-free** and available at sampling time from a symmetry template, exactly as DiffCSP++/WyckoffDiff-style generators supply Wyckoff templates for atoms^{8,9}. This yields 238 operation-cosets across the corpus (versus 707 for a codebook clustered from the target rotations, whose index is derived in part from the answer and so serves only as an upper bound). The coset index enters the network as a per-molecule embedding and is threaded through the sampler, conditioning both training and generation.

SO(3)-averaged objective

The SO(3) flow-matching residual has high variance because the prior rotation is drawn uniformly. Following the SO(3)-averaged flow objective for conformer generation¹⁶, we average the orientation residual over K prior draws per molecule before the gradient step, reducing target variance at fixed data.

Template-free coset prediction

For template-free deployment the coset must be inferred. We train a classifier that predicts each molecule’s coset from the conformer, centroid, lattice, and space group—*without* orientation, so

it cannot leak the target. Because its argmax is noisy, we *marginalize the top-k*: condition the flow on each of the k most-likely cosets and select the best draw (in deployment by R_{wp}/energy ; here, an oracle upper bound).

Full symmetry conditioning: the crystal-family lattice mask

The coset conditions orientation; the lattice remains free. Molecular crystals are overwhelmingly monoclinic and orthorhombic, so reference cells place $\sim 74\%$ of cell angles at exactly 90° , yet an unconstrained 9-DOF shape head reproduces this for only $\sim 16\%$. We condition the lattice on the crystal family the way DiffCSP++⁸ conditions inorganic sites: represent the cell by the $O(3)$ -invariant log-metric $\log(L^T L) \in \mathbb{R}^6$, whose first coordinate is the volume and whose remaining coordinates are the deviatoric shape directions, and apply a binary *family mask* that freezes the constrained coordinates to their canonical values after each sampler step and in the prior (basis and per-family mask in the Supporting Information). The flow thus learns only the free, family-consistent degrees of freedom. The mask mechanism is DiffCSP++’s; our contribution is transporting it into the molecular rigid-body flow and composing it with the orientation coset, giving a fully symmetry-conditioned flow.

Exact-match finishing: physical press, learned prior, self-conditioning, recycling

Symmetry conditioning is necessary but not sufficient for exact match; a small stack of levers supplies the remaining positioning precision, each ablated independently (§2.8). (i) *Rigid-press finisher*: an unsupervised, symmetry- and rigidity-preserving relaxation of the generated packing—a Lennard-Jones energy over intermolecular atom pairs (van der Waals radii), optimized over the asymmetric-unit pose and the family-masked cell, with symmetry copies regenerated from the space-group operations each step, so it cannot break the imposed symmetry and never sees the target. (ii) *Learned centroid prior*: optimal-transport coupling¹⁷ with a fixed

Gaussian centroid prior repairs an under-trained centroid head. (iii) *Self-conditioning*: the model optionally consumes its own terminal estimate of the structure, trained with probability 0.5 and carried across sampler steps. (iv) *Recycling*: AlphaFold-style, the sampling trajectory is re-solved conditioned on the previous pass’s output structure. (v) *Orientation test-time augmentation (TTA)*: perturbing the asymmetric-unit orientation and pooling the best finished draw. All are gated behind configuration flags, leaving the default generation path unchanged (finisher energy/optimizer and the self-conditioning update in the Supporting Information).

Data and evaluation

The corpus is a seeded CSD sample¹²: 1,127 ordered organic molecular crystals (5,048 rigid blocks, a fixed 964/131 split), with 1,095 crystals containing ≥ 2 copies of a species—the multi-copy subset where a relative target exists—and a 2,539-crystal scale corpus for §2.6. The primary training metric is the held-out non-reference orientation-loss drop (percentage reduction from untrained to trained on non-reference copies, which cannot be earned by predicting the identity). We also report orientation-isolated and end-to-end STRUCTUREMATCHER reconstruction¹⁸ with best-of- k sampling, following match-rate cautions¹⁴. All learning numbers are means over three seeds.

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Data and Software Availability

The SymMC-Flow source code, the CSD refcode manifest, and the export and factorization pipeline are openly available at <https://github.com/fronkt/symmc-flow> and archived at Zenodo¹⁹.

The Cambridge Structural Database is licence-restricted; the underlying crystal structures are not redistributed, but the deposited refcode manifest reproduces the corpus from a licensed CSD installation. The Supporting Information document contains the extended methods.

Associated Content

Supporting Information.

Rigid-body factorization, intrinsic gauge, manifold parametrizations, and deployable coset recovery; the R_{asym} conditional probe; the crystal-family lattice mask (log-metric basis and per-family free/frozen coordinates); the rigid-press finisher energy and optimizer and the self-conditioning and recycling update rules; and per-seed numbers with the tolerance sweep (PDF).

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